

# P1 consolidation

A mixed checkpoint from quadratics to calculus

## YOUR AIM TODAY

**Retrieve methods without topic labels and diagnose the next weak area.**

### YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: \_\_\_\_\_ Finish time: \_\_\_\_\_

Date: \_\_\_\_\_

# The ideas you need

Mixed practice trains the hardest exam skill: selecting the method yourself.

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- Read the command word and write the topic beside each question only after attempting it.
- If stuck, write the relevant formula before doing arithmetic.
- Check algebra, interval, units and exact form.
- Target: at least 5 of 6 fully correct.

## MEMORY RULE

Choose method, show structure, calculate, check.

## Model selection

### WORKED STEPS

A tangent question needs differentiation; an arc needs radians; a repeated root needs discriminant zero.

## Repair rule

### WORKED STEPS

Do not erase an error. Mark the first wrong line, state the rule, then redo the whole question cleanly.

# Independent practice

Show every step. Include units in Mechanics.

| Question                                                   | Working and final answer |
|------------------------------------------------------------|--------------------------|
| 1. Solve $x^2 - 7x + 10 = 0$ .                             |                          |
| 2. Find inverse of $f(x) = 3x - 4$ .                       |                          |
| 3. Find line through (2,1) perpendicular to $y = 2x + 3$ . |                          |
| 4. Find coefficient of $x^2$ in $(1 + 2x)^4$ .             |                          |
| 5. Find stationary point of $y = x^2 - 6x + 4$ .           |                          |
| 6. Evaluate integral 0 to 2 of $(3x^2 + 2) \, dx$ .        |                          |

**STOP**

Do not continue until all six questions have an attempt.

# Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

| Q | Answer and essential working                          |
|---|-------------------------------------------------------|
| 1 | $(x-5)(x-2)=0$ , so $x=2$ or $5$ .                    |
| 2 | $f^{-1}(x)=(x+4)/3$ .                                 |
| 3 | Gradient $-1/2$ ; $y-1=-(1/2)(x-2)$ , so $y=-x/2+2$ . |
| 4 | $C(4,2)2^2=24$ .                                      |
| 5 | $dy/dx=2x-6=0$ gives $x=3$ ; $y=-5$ . Minimum.        |
| 6 | $[x^3+2x]_0^2=8+4=12$ .                               |

| Q | First wrong step | Correct rule |
|---|------------------|--------------|
|   |                  |              |
|   |                  |              |
|   |                  |              |

## SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.